

Rajnish Kumar

Generative AI Engineer | LLMs | RAG | AI Agents | LangChain | LangGraph

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Professional Summary

Generative AI Engineer with 2+ years of experience building enterprise AI solutions using Python, Gemini, Google Vertex AI, RAG, LangChain, and LangGraph. Experienced in designing agentic workflows, RAG pipelines, AI assistants, and multi-step LLM applications with retrieval, tool integration, vector databases, knowledge graphs, memory, and structured outputs. Hands-on experience developing FastAPI-based AI services, integrating LLM APIs, and building production-oriented GenAI applications with Docker and CI/CD.

Professional Experience

LTIMindtree

Jun 2024 – Sep 2026

Software Engineer — Generative AI

- Developed enterprise GenAI solutions for banking use cases using Python, Google Vertex AI, Gemini, RAG, LangChain, and LangGraph.
- Built a **GenAI Operations & Incident Assistant** using Gemini and RAG to retrieve relevant operational knowledge, analyze incident context, summarize issues, and generate troubleshooting recommendations.
- Developed an **AI Event & Incident Triage Agent** using LangGraph-based agentic workflows to analyze incoming events, retrieve historical incidents and technical documentation, and generate incident analysis and recommended actions.
- Designed RAG pipelines covering document ingestion, text chunking, **Vertex AI Embeddings**, semantic retrieval, vector search, context construction, and Gemini-based response generation.
- Implemented LangGraph-based workflows with state management, conditional routing, tool integration, and multi-step LLM orchestration for enterprise AI applications.
- Developed Python-based backend services and REST APIs using FastAPI, implementing structured outputs, response validation, error handling, and performance optimization for GenAI applications.

LTIMindtree

Feb 2023 – Apr 2023

Cloud Engineer Intern

- Assisted in provisioning AWS cloud infrastructure and deploying containerized applications using Docker.
- Supported deployment automation, infrastructure monitoring, cloud operations, and infrastructure management activities.

Technical Skills

Programming: Python, SQL

Generative AI: LLMs, RAG, AI Agents, Agentic AI, Multi-Agent Systems, Prompt Engineering, Tool Calling, Function Calling

LLM & AI Platforms: Gemini, Google Vertex AI, OpenAI API, LangChain, LangGraph, HuggingFace Transformers, Sentence Transformers

RAG & Vector Search: RAG, Knowledge Graphs, Graph-based Retrieval, Embeddings, Vertex AI Embeddings, Semantic Search, Document Ingestion, Text Chunking, Similarity Search, Top-K Retrieval, ChromaDB

AI Engineering: LLM Orchestration, Agent Workflows, Memory, State Management, API Integration, Response Validation, Caching

Backend & DevOps: FastAPI, REST APIs, Pydantic, Docker, AWS EC2, GitHub Actions, CI/CD

Tools: Git, GitHub, Linux

Projects

Production-Ready Multi-Agent AI Trading Research Agent

GitHub Repo

Python, LangGraph, LangChain, ChromaDB, Knowledge Graphs, Gemini, Groq, Cerebras, Docker, AWS

- Designed and developed a production-ready Multi-Agent AI Trading Research Agent using LangGraph, RAG, and LLMs to automate financial market research and report generation.
- Architected modular Retriever, News, Sentiment, Risk, Summary, and Memory agents using ChromaDB, Hugging-Face Embeddings, and Knowledge Graphs for context-aware analysis.
- Integrated Tavily Search, yFinance, and multiple LLM providers (Gemini, Groq, Cerebras) for real-time market data, news analysis, and semantic retrieval.
- Enhanced the RAG pipeline with graph-based retrieval to model relationships between financial entities, market events, news, and concepts for improved multi-hop contextual reasoning.
- Developed FastAPI REST APIs with health checks, Docker containerization, and automated CI/CD deployment to AWS EC2 using GitHub Actions and Docker Hub.
- Engineered reliable deployments using semantic versioning, idempotent deployment scripts, retry-based health checks, and Docker restart policies.
- Optimized performance through embedding reuse, persistent vector storage, graph-based retrieval, and efficient semantic search.

HireSense AI – LLM-Based Resume Intelligence & Ranking Engine

GitHub Repo

Python, OpenAI API, LangChain, FastAPI, Streamlit, Embeddings

- Built an LLM-powered platform for semantic resume-job matching, candidate evaluation, and intelligent resume ranking.
- Developed embedding-based semantic similarity pipelines for resume and job-description matching.
- Implemented AI-driven skill extraction, resume evaluation, and contextual feedback generation using OpenAI models.
- Developed FastAPI REST APIs and optimized AI workflows using embedding reuse, response caching, and retrieval optimization.

Image2Insight – OCR-Based Sentiment Analysis

GitHub Repo

Python, OCR, NLP, VADER Sentiment Analysis

- Developed an OCR-based NLP application to extract textual information from image documents and generate automated insights.
- Implemented image preprocessing and text-processing pipelines to improve OCR output and downstream analysis.
- Applied VADER sentiment analysis to classify extracted text and automate sentiment-based insights.

Leadership & Communication

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- Delivered technical presentations and communicated complex technical concepts to diverse audiences.
 - Collaborated with cross-functional teams to drive knowledge sharing and achieve common objectives.
 - Coordinated team initiatives and supported peer mentoring while strengthening stakeholder communication, active listening, problem-solving, and feedback skills.

Education

Global Academy of Technology

Bengaluru, India

Bachelor of Engineering (B.E.)

Electronics and Communication Engineering

2019 – 2023